International Islamic University Chittagong
Center for General Education (CGED)
Semester End Examination Autumn-2025
Course Code: GEBL-2401
Course Title: Bangla Language and Literature
Full Marks: 50 Time: 2.5 Hours

ক-বিভাগ
ভাষা ও নির্মিতি: ৩০
(ডান পাশের সংখ্যা প্রশ্নের মান জ্ঞাপক)

প্রশ্ন নং	বর্ণনা	মান	CLO	Cognitive learning
০১.	ক. প্রমিত বাংলা বানানের পাঁচটি নিয়ম উদাহরণসহ লেখ।	০৫	CLO1	Understand
	খ.কমার পাঁচটি ব্যবহার আলোচনা কর।	০৫	CLO1	Understand
০২.	পত্রিকায় প্রকাশের উপযোগী **"বাংলাদেশের গণতান্ত্রিক অভিযাত্রায় গণভোট ২০২৬ এর ভূমিকা"** – শীর্ষক একটি পত্র রচনা কর।	১০	CLO3	Create
	অথবা			
	একটি স্বরচিত খুদে গল্প উপস্থাপন কর।	১০	CLO3	Create
০৩.	**সংক্ষেপে আলোচনা কর:** ক. মানবতা ও নৈতিকতা। খ. আধুনিক তথ্য-প্রযুক্তি।	৫×২=১০	CLO3	Create

খ-বিভাগ
সাহিত্য: ২০

প্রশ্ন নং	বর্ণনা	মান	CLO	Cognitive learning
০৪.	**"বঙ্গভাষা"** কবিতার আলোকে কবি মাইকেল মধুসূদন দত্তের মাতৃভাষা প্রেমের পরিচয় দাও।	১০	CLO3	Analyze
০৫.	**"কবর"** নাটকে বাঙালির প্রতিবাদী চেতনার প্রতীক হিসেবে মুর্দা ফকির চরিত্র বিশ্লেষণ কর ।	১০	CLO3	Evaluate

সমাপ্ত

Bismillahir Rahmanir Rahim

International Islamic University Chittagong

Department of Computer Science & Engineering

B. Sc. in CSE Semester Final Examination, Autumn-2025

Course Code: CSE2423 Course Title: Database Management System

Total marks: ***50*** Time: ***2*** hours ***30*** minutes

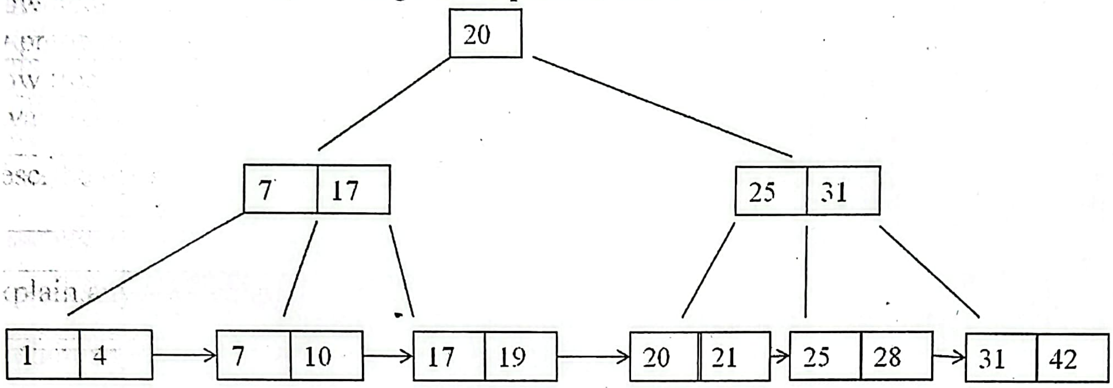
[Answer ***all*** the questions. Figures in the right hand margin indicate full marks. Separate answer script must be used for Group A and Group B]

Group A

Q. No.	Question	CLO	BT	Marks
1.a	Differentiate between domain constraint and integrity constraint with appropriate example.	2	A	4
1.b	Suppose, you are implementing a hospital management system with schema: Doctor (<u>docID</u>, dName, specialization, phone, salary) Patient (<u>patientID</u>, patientName, gender, age, nid, <u>docID</u>) Appointment (<u>appointID</u>, appointDate, <u>patientID</u>, <u>docID</u>) Bill (<u>billID</u>, <u>patientID</u>, totalAmount, paymentStatus) Write SQL DDL statement for creating these tables including all domain and integrity constraints. Make sure to maintain following condition regarding constraints: i. dName, specialization, patientName, appointDate must not be NULL ii. salary must be greater than 20000 iii. age must be greater than 0. iv. gender will be 'M' for male, 'F' for female v. paymentStatus will be 'Paid' or 'Unpaid' vi. phone, nid must be unique vii. Primary Key and Foreign Key must be included.	2	A	6
	OR			
1.b	Why trigger is used in DBMS? Suppose, you want to keep a log table after the update of payment of bill for patient. Use trigger to implement the log table where you can keep track the patient who have paid their bill. You can create different log table for patient.			
2.a	Consider relation R(WXYZPQ) with the following functional dependencies: $W \rightarrow X$ $X \rightarrow Y$ $ZP \rightarrow W$ $Q \rightarrow Z$ i) Identify all candidate keys. ii) Decompose the relation into BCNF with justification.	2	A	6
2.b	Explain lossless-join decomposition and dependency preservation. Show why both properties are important using examples.	2	A	4
	OR			
2.b	Given R(A, B, C, D, E): $F1 = \{A \rightarrow BC, C \rightarrow D, DE \rightarrow A\}$ $F2 = \{A \rightarrow BCD, E \rightarrow A\}$ Determine whether F1 and F2 are equivalent.			

Group B

Q. No.	Question	CLO	BT	Marks
3.a	Consider the key values: 5, 7, 12, 15, 20, 25, 29. Check whether there will be overflow if you apply hash function, h(K) = K % 2 and each bucket with maximum 2 keys in static hashing. If there are overflow in bucket, how will you solve it?	2	A	4
	OR			
3.a	For the same keys given above, check whether there will be bucket overflow in dynamic hashing if bucket capacity is 2 and global depth is 2. If there are overflow in bucket, handle the bucket overflow using dynamic hashing method.			
3.b	Suppose you are given a 4[th] order B+ tree in following diagram. Through step-by-step deduction of keys, K = {21, 31, 20, 10, 7, 25, 42, 4} from B+ tree, find out the final B+ tree after deletion of keys with given sequential order in set.	2	A	6



Q. No.	Question	CLO	BT	Marks
4.a	Discuss the problems that may arise if concurrency control is not applied; Give example	2	A	4
4.b	Analyze the concept of serializability and its types with suitable examples.	2	A	6
	OR			
4.b	Critically analyze the concept of *Dirty read*, *Fuzzy read*, and *Phantom* with example			
5.a	How does deadlock encounter in lock based protocol? How you will handle deadlock by preemptive and non-preemptive approach.	2	U	4
5.b	How tree based protocol in concurrency control can ensure consistency? What are the advantages and disadvantages of tree based protocol.	2	U	3
5.c	Describe the purpose of checkpoints in database recovery mechanisms.	2	U	3
	OR			
5.c	Explain any two techniques used to handle transaction failures in DBMS.	2	U	3

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE Final Examination, Autumn 2025

Course Code: CSE-2427 Theory of Computation

Total marks: 50 *Time: 2 hours 30 minutes*

[Figures in the right-hand margin indicate full marks

Course Outcomes and Bloom's taxonomy levels are mentioned in additional columns]

Group-A

		CO	DL	
1. a)	Differentiate between a Context-Free Grammar (CFG) and a Regular Grammar. **OR** Define **left-most derivation** and **right-most derivation** with a brief example.	CO1	R	3
b)	What is **ambiguity**? Show that the following grammar is ambiguous by showing two distinct parse trees (or derivations) for the string **aab**. $S \rightarrow aS \mid Sa \mid b$	CO1	N	3
c)	Consider the following grammar: $S \rightarrow 0B \mid 1A$ $A \rightarrow 0 \mid 0S \mid 1AA$ $B \rightarrow 1 \mid 1S \mid 0BB$ Derive the string 001101 using a **right-most derivation**. Draw the parse tree for this string. **OR** Derive the string 001101 using a **left-most derivation**. Draw the parse tree for this string.	CO2	A	4
2. a)	Convert any one of the following CFGs into an equivalent CFG in *Chomsky normal form*. $S \rightarrow aSb \mid AB \mid \varepsilon$ $A \rightarrow aB$ $B \rightarrow Ab$ **OR** $S \rightarrow ASA \mid aB$ $A \rightarrow B \mid S$ $B \rightarrow b \mid \varepsilon$	CO2	A	3
b)	If it is possible, give a context-free grammar (CFG) for the following languages over the alphabet $\Sigma = \{0,1\}$. All strings in the language **L:** $\{ww \mid w \in \{0,1\}^*\}$ Or else, establish with a proper technical proof that it is not possible.	CO1	C	3
c)	Give a context-free grammar (CFG) for each of the following languages over the alphabet $\Sigma = \{0,1\}$. - All strings in the language **L:** $\{0^n1^{2n} \mid n \geq 1\}$ - All strings with an unequal number of 0s and 1s.	CO2	C	4

Group-B

		CO	Level	Marks
3.				
a)	Let a language L is defined as L = {(M,w) | M is a TM and M halts on input w}. Show that L is undecidable.	**CO3**	**N**	**3**
b)	Convert the following context-free grammar (CFG) to an equivalent pushdown automaton $S \rightarrow aXbX$ $X \rightarrow aY \mid bY \mid \varepsilon$ $Y \rightarrow X \mid c$	**CO1**	**N**	**3**
c)	Construct a pushdown automaton that recognizes a language accepting **all strings of balanced parentheses.** **OR** Construct a pushdown automaton that recognizes all strings with an **unequal number of 0s and 1s** over the alphabet **Σ ={0,1}.**	**CO3**	**C**	**4**
4.				
a)	a) Consider a Turing Machine with the transition function δ defined as: $\delta(q_0, 0) = (q_1, 1, R)$ $\delta(q_1, 1) = (q_0, 0, R)$ $\delta(q_1, \_) = (q_{accept}, \_, R)$ The symbol _ means blank. Give the sequence of **configurations** that the machine enters when started with the input string 010.	**CO3**	**N**	**3**
b)	Let the language A_{DFA} be described as A_{DFA} = {(B,w)| B is a DFA that accepts the input string w}. Prove that A_{DFA} is a decidable language. **OR** Let the language A_{NFA} be described as A_{NFA} = {(N,w)| N is a NFA that accepts the input string w}. Prove that A_{NFA} is a decidable language.	**CO4**	**E**	**3**
c)	Give the implementation-level description of a Turing machine that decides the following languages. $L = \{a^n b^{2n} c^{4n} \mid n \geq 0\}$	**CO3**	**C**	**4**
5.				
a)	State the SUBSET-SUM problem formally. Sketch briefly the outline to prove that it is in NP. **OR** What is polynomial-time reducibility? Briefly explain how it is relevant in complexity theoretic study.	**CO5**	**U**	**3**
b)	Explain the technical difference between the complexity classes **NP-Hard** and **NP-Complete.**	**CO5**	**U**	**3**
c)	Prove that the set of **Real Numbers** is uncountable using the diagonalization method.	**CO4**	**N**	**4**

International Islamic University Chittagong

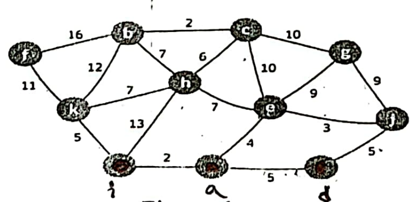
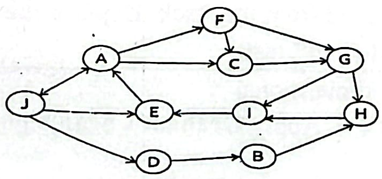
Department of Computer Science and Engineering

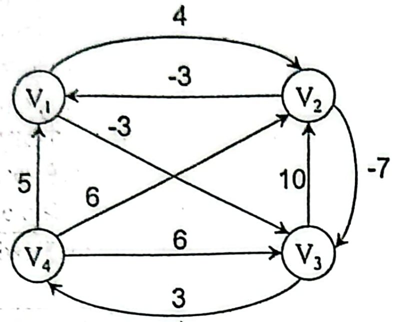
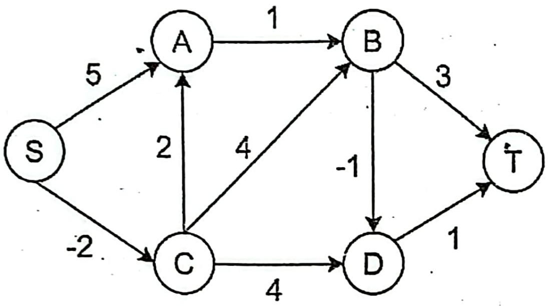
B.Sc. in CSE, Final Examination Autumn 2025

Course Code: CSE-2421, Course Title: Computer Algorithm

Total Marks: 50, Time: 2 hour 30 minutes

The figures in the right margin show the full marks. Use separate answer scripts for Group A and Group B.

	Group-A	CO	DL	Marks
1.	a) A greedy algorithm makes a locally optimal choice at each step. Explain why the greedy-choice property alone is not sufficient to guarantee a globally optimal solution. Support your answer with reasoning. **Or** a) Two students propose greedy rules for Activity Selection: Student A: choose the activity with the shortest duration Student B: choose the activity with the earliest finishing time. Analyze which rule is correct and justify your answer logically.	CO4	C4	3
	b) Both Fractional Knapsack and 0/1 Knapsack exhibit optimal substructure. Explain analytically why greedy succeeds in one but fails in the other.	CO4	C4	3
	c) Construct the optimal Huffman codes for these symbols based on their occurrence frequencies. **Symbol**: A, B, C, D, E, F **Frequency**: 4, 7, 11, 18, 29, 47 Show steps of the construction of the Huffman tree.	CO4	C3	4
2.	a) BFS and DFS both have time complexity O(V+E). Explain why BFS is still preferred for shortest path problems in unweighted graphs.	CO3	C4	3
	b) Run Kruskal's algorithm to find a minimum spanning tree of the graph in Figure: 1. **Or** b) Kruskal's algorithm explicitly checks for cycles before adding an edge. Analyze why cycle detection is necessary in Kruskal's algorithm but not in Prim's.	CO3	C4	3
	c) Using Prim's Algorithm, construct the Minimum Spanning Tree (MST) of the graph, starting from vertex b.  Figure: 1 **Or** c) Consider the following graph, Figure: 2. In what order will the nodes be visited using a Breadth First Search? In what order will the nodes be visited using a Depth First Search? (Assuming B is the source, show the state of the queue and stack in case of BFS and DFS respectively.)  Figure:2	CO3	C3	4

	Group-B	CO	DL	Marks
3.	a) Consider the following weighted directed graph for finding the all-pairs shortest paths using the Floyd–Warshall algorithm.  $D^{(0)} = \begin{bmatrix} 0 & 4 & -3 & \infty \\ -3 & 0 & -7 & \infty \\ \infty & 10 & 0 & 3 \\ 5 & 6 & 6 & 0 \end{bmatrix}$ Figure: 3 Compute the matrix $D^{(1)}$ and $D^{(2)}$ applying the **Floyd–Warshall algorithm.** **Or** a)  Figure: 4 i) Using vertex S as the source, apply the **Bellman–Ford algorithm** to compute the shortest path distances from S all other vertices. ii) Show the distance values **after each iteration** of the outer loop is completed.	CO3	C4	4
	b) "Subpaths of shortest paths are also shortest path" justify your answer with a suitable example.	CO3	C4	3
	c) Consider a graph with one negative edge but no negative cycle. Explain why Dijkstra's algorithm may still give incorrect results.	CO3	C3	3
	1			
4.	a) A bee is flying from point A(2,3)to B(8,6)in a straight line. From point B, it then flies toward point C(6,1). i) Determine whether the bee makes a **left turn or a right turn** at point B. ii) Justify your answer using the **cross-product technique.**	CO4	C3	4
	b) Find the GCD of the following integers using **Euclid's algorithm** gcd(540,168) **Or** b) During Graham's Scan, some points are removed from the stack. **Explain the geometric meaning of removing a point using the "right-turn" test.**	CO4	C4	3
	c) Consider the following set of points in a two-dimensional plane:{(1,2), (3,1), (5,3), (4,6), (2,5), (6,4), (3,4)} Apply **Graham's Scan algorithm** to determine the **convex hull** of the given points.	CO4	C3	3

5.	a) Explain why optimization versions of NP-complete problems (e.g., Traveling Salesman Optimization) are NP-hard, while their decision versions are NP-complete.	CO2	C4	3
	b) Describe the backtracking approach to solving the N-Queens problem. Show the steps of a backtracking algorithm for placing 4 queens in 4×4 chessboard. **Or** b) A steadfast delivery man is assigned to deliver important products across the busy areas of **Chawkbazar(A)**, **Kotowali(B)**, **Hathazari(C)**, **2 No. Gate(D)**, and **Muradpur(E)**. To ensure delivery time and minimize travel distance, he must start from one location, visit each area exactly once, and return to the starting point by choosing the most efficient route. A is the starting point.	CO4	C4	3
	c) What technique does the branch and bound technique apply to reduce the search space? Consider the following scenario of five cities: A, B, C, D, E. Distance between the cities are as follows. A-B: 10, A-C: 15, A-D: 20, A-E: 25, B-C: 35, B-D: 30, B-E: 20, C-D: 18, C-E: 22, and D-E: 12. You have to start from any random city, visit each city exactly once and come back to the starting city with minimum cost. Show how a branch and bound algorithm explores the search space in this scenario.	CO4	C3	4

Distance matrix for 5(b) (Or):

	A	*B*	*C*	*D*	*E*
A	0	12	10	19	8
B	12	0	3	7	6
C	10	3	0	2	20
D	19	7	2	0	4
E	8	6	20	4	0

International Islamic University Chittagong (IIUC)
Department of Computer Science and Engineering (CSE)
Semester End Examination

Program: B. Sc. in CSE	**Semester: Autumn-2025**
Course Code: MATH-2407	**Course Title: Mathematics-IV**
Time: 2.5 hours	**Total Marks: 50**

(i) Answer all the questions. The figures in the right-hand margin indicate full marks.
(ii) Separate answer script must be used for separate group.
(iii) Course Learning Outcomes (CLOs) and Bloom's Levels are mentioned in additional Columns.

Course Learning Outcomes (CLOs) of the Questions

CLO1	Demonstrate the understanding of the basic principles and operations set theory, complex numbers, geometrical interpretation complex functions and the concept of transformation in a complex plane. different types of functions and signals
CLO2	Apply the concept of transformation of an object into complex space and operation of complex functions
CLO3	UseFourier series, Laplace's Transforms, Fourier Transform in different scenario.
CLO4	Analyze the harmonics & spectrum of different types of waves.
CLO5	Demonstrate the harmonic analysis using MATLAB.

Bloom's Taxonomy Domain Levels of the Questions

Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Group-A

		Mark	CLO	DL
1.	i. Sketch 3 cycles for the function represented by $$y = f(t) = \begin{cases} 0; & -5 \le t < 0 \\ 4; & 0 \le t < 5 \end{cases} \qquad T = 2L = 10 \quad \therefore L = 5$$ ii. Find the Fourier series for the above function.	2+8	CLO3	App
	Or Derive the complex form of Fourier series.	10	CLO3	App
2. a)	Evaluate the convolution sums of $y[n] = x[n] * h[n]$ Where, $$x[n] = \begin{cases} 1, & n = 0 \\ 3, & n = 1 \end{cases} \; and \; h[n] = \begin{cases} 2, & n = 0 \\ 1, & n = 1 \end{cases}$$ Where, n represents the time index.	6	CLO4	U
b)	How would you explain the functioning of the cochlea inside the ear as a practical example of a Fourier series?	2	CLO4	U
c)	Plot the line (**at least 4**) spectrum for the following complex wave $$f(t) = 2\pi + \sum_{n=1}^{500} \frac{2}{n} \sin n\pi t$$	2	CLO4	An

Group-B

			Mark	CLO	DL
3.	a)	Find Fourier Transform of $f(t) = 1 \quad ; 0 \le t < 1$ $= -1 \quad ; -1 \le t < 0$ $= 0 \quad ; \lvert t \rvert > 1$ **Or** Find the inverse Laplace transform of $\dfrac{s+3}{s(s-1)(s-2)}$	5 5	CLO3 CLO3	App App
	b)	Why is the line spectrum continuous in the Fourier transform?	2	CLO4	U
	c)	What type of signals do Fourier series, Fourier transform, and Laplace transform analyze?	2	CLO3	U
	d)	Give an example of an unstable signal (function)	1	CLO3	U
4	a)	Solve the following Initial Value Problem (IVP) by Laplace Transform $Y'' - 3Y' + 2Y = 4e^{2t} \quad Y(0) = -3 \qquad Y'(0) = 5$ **Or** Evaluate $\mathcal{L}\{t^2.\sin 3t\}$ using Multiplication theorem.	5 5	CLO3 CLO3	App App
	b)	Draw the graph for the following function: $x(t) = -u(t+3) + 2u(t+1) - 2u(t-1) + u(t-3)$	5	CLO1	U
5.	a)	Write down a user defined function in MATLAB to reconstruct $f(t)$ in the time interval of **[-4, 20]** for the following complex wave and **also write MATLAB code to show line Spectrum for the following wave in one window** $f(t) = \dfrac{\pi}{3} + \sum_{n=1}^{500} \dfrac{5}{n} \sin n\pi t$	8	CLO5	An
	b)	If $x[n] = 4 \quad ; n = 0$ and $h[n] = 3 \quad ; n = 0$ $= 5 \quad ; n = 1$ $\qquad = 6 \quad ; n = 1$ Write MATLAB code to find the convolution sum of the above signals.	2	CLO5	An

International Islamic University Chittagong
Department of Computer Science and Engineering
B. Sc. Engineering in CSE, **Semester End Exam**, Autumn 2025

Course Code: **ACC 2401** Course Title: **Financial & Managerial Accounting**
Time: 2 hours 30 minutes Full Marks: 50

(i) The figures in the right-hand margin indicate full marks
(ii) Course Outcomes and Bloom's Levels are mentioned in additional Columns

Part A

[Answer the questions from the followings]

		CO	DL	
1.a	The unadjusted trial balance of Joe Heider, Attorney at July 31, 2025, and the related month-end adjustment data are as follows:	CO2	C2	10

Joe Heider
Trial Balance
July 31, 2025

	Amount ($)	Amount ($)
Cash	14,600	
Accounts Receivable	11,600	
Prepaid Rent	3,600	
Supplies	800	
Furniture	16,800	
Accumulated Depreciation		3,500
Accounts Payable		3,450
Salary Payable		-
Joe Heider Capital		38,650
Joe Heider Withdrawals	4,000	-
Legal Service Revenue		8,750
Salary Expense	2,400	
Rent Expense	-	
Utilities Expense	550	
Depreciation Expense		
Supplies Expense		
	54,350	54,350

Adjustment data: a) Prepaid rent expired during the month. The unadjusted prepaid balance of $3,600 relates to the period July through October. b) Supplies on hand at July 31, $500. c) Depreciation on furniture for the month. The estimated useful life of the furniture is four years. d) Accrued salary expense at July 31 for one day only. The five-day weekly payroll is $1,000. e) Accrued legal service revenue at July 31, $700.

Required:

i) Prepare the adjusted trial balance.
ii) Prepare the Owners' Equity statement.

2.a **K.M. Atorney** CO2

Adjusted Trial Balance as at 31 December 2023

Particular	Dr($)	Cr($)
Cash	90,000	
Accounts receivable	20,000	
Interest Receivable	200	
Notes Receivable	4,000	
Prepaid Insurance	960	
Prepaid Rent	2,400	
Supplies	1,520	
Equipment	Last Four digit of student ID	
Depreciation- Equipment		12,500
Buildings	140000	
Depreciation- Buildings		15,000
Land	56,240	
Accounts payable		Last Four digit of student ID
Notes Payable		10,920
Tax Payable		10,000
Interest payable		750
Salaries payable		7,000
Unearned revenue		20,200
Capital		100000
Drawings	40,000	
Service revenue		360000
Advertising expense	1200	
Salary expense	150000	
Rental expense	9,600	
Supplies expense	2,280	
Office expense	10,000	
Interest expense	750	
Insurance expense	1,920	
Dep. expense- Equipment	2,500	
Dep. expense- Building	3,000	
Interest revenue		200
Total	536,570	536570

Required: Financial statements of **K.M. Atorney** for the year ended 31 December 2023.

Or,

The trial balance of Infinia Group at May 31, 2025, follows: **CO2 C3 10**

Sl. No.	Name of Accounts	Ref.	Debit (Tk.)	Credit (Tk.)
1	Cash		1,670	-
2	Notes Receivable		10,340	-
3	Interest Receivable		-	-
4	Supplies		560	-
5	Prepaid Insurance		1,790	-
6	Furniture		27,410	-

7	Accumulated Depreciation-Furniture		-	1,480
8	Building		55,900	-
9	Accumulated Depreciation-Building		-	33,560
10	Land		13,700	
11	Accounts Payable		-	14,730
12	Interest Payable		-	-
13	Salary Payable		-	-
14	Unearned Service Revenue		-	6,800
15	Notes Payable-Long Term		-	18,700
16	Capital		-	34,290
17	Withdrawal		3,800	-
18	Service Revenue		-	9,970
19	Interest Revenue		-	-
20	Depreciation Expense-Furniture		-	-
21	Depreciation Expense-Building		-	-
22	Salary Expense		2,170	-
23	Interest Expense		-	-
24	Utilities Expense		490	-
25	Property Tax Expense		640	
26	Advertisement Expense		1,060	-
27	Supplies Expense		-	-
	Total		119,530	119,530

Additional data:

i) Accrued salary expense Tk. 600
ii) Supplies in hand Tk. 410
iii) Prepaid insurance expired during May, Tk. 390
iv) Accrued interest expense, Tk. 220
v) Unearned service revenue earned during May, Tk. 4,400
vi) Accrued advertising expense, Tk. 60 (Credit accounts payable)
vii) Accrued interest revenue, 170
viii) Depreciation: furniture-Tk. 380; building-Tk. 160.

Required: Complete Infinia's worksheet for May 31, 2025.

Part B

[Answer the questions from the followings]

3.a The following data are from the accounts of BSA Group: CO2 **C3** 10

Inventories	July 1, 2023	June 30, 2024
Finished Goods	Tk. 20,000	Tk. 28,000
Work in Process	60,000	36,000
Materials	40,000	48,000

Sales Discounts	Tk. 8,000
Purchase Discounts	3,200
Sales	1,800,000
Purchase Returns and Allowances	20,000
Depreciation-Factory Machinery	160,000
Factory Insurance	50,000
Freight Out	8,000
Other Factory Expenses	16,000
Bond Interest Expense	50,000

Sales Salaries	100,000
Freight In	12,000
Direct Factory Labor	800,000
Material Purchases	400,000
Advertising Expenses	12,000

Required: Prepare a Cost of Goods Manufactured Statement for the year ended June 30, 2024.

4.a Superior Door Company sells prehung doors to home builders. The doors are sold for \$60 each. Variable costs are \$32 per door, and fixed costs total \$450,000 per year. The company is currently selling 30,000 doors per year. CO3 C3 10

Required:

i) Prepare a contribution format income statement for the company at the present level of sales and compute the degree of operating leverage.

ii) Management is confident that the company can sell 37,500 doors next year (a year of 7,500 doors, or 25%, over current sales). Compute the following:

a) The expected percentage increase in net operating income for next year.

b) The expected total dollar net operating income for next year. (Do not prepare an income statement; use the degree of operating leverage to compute your answer).

OR

Super Sales Company is the exclusive distributor for revolutionary bookbag. The product sells for \$50 per unit and has a CM ratio of 40%. The company's fixed expenses are \$360,000 per year. CO3 C3 10

Required:

i) What are the variable expenses per unit?

ii) Using the equation method:

a) What is the break-even point in units and in sales dollars?

b) What sales level in units and in sales dollars is required to earn an annual profit of \$90,000?

c) Assume that through negotiation with the manufacturer the Super Sales Company is able to reduce its variable expense by \$3 per unit. What is the company's new break-even point in units and in sales dollars?

5.a Peak sales for Midwest Products, Inc. occur in August. The company's sales budget for the third quarter showing these peak sales is given below: CO1 C4 10

	July	August	September	Total
Budgeted sales	\$600,000	\$900,000	\$500,000	\$2,000,000

From past experience, the company has learned that 20% of a month's sales are collected in the month of sale, that another 70% is collected in the month of following sale, and that the remaining 10% is collected in the second month following sale. Bad debts are negligible and can be ignored. May sales totaled \$430,000, and June sales totaled \$540,000.

Required:

a. Prepare a schedule of expected cash collections from sales. by month and in total, for the third quarter.

b. Assume that the company will prepare a budgeted balance sheet as of September 30. Compute the accounts receivable as of that date.